Mustafa Mehboob

Product Engineer · Security · Payments · Cloud

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Summary

Full-stack engineer with **3+ years** on products that take payments and hold personal data, across visa services, e-commerce, analytics and industrial control. Took over UKVisaPortal during a live cyber attack and **led its three-engineer team as revenue passed $1M**, co-led WorldVisaHub, led Userbird (now LinklyHQ) and mDrive Asia, and build the backend of a live SCADA platform. **First author of a 2026 paper on LLM-based software testing and vulnerability detection**.

Experience

Software Engineer Jan 2026 – Present
Softaims *Lahore, Pakistan*

- **Took over UKVisaPortal during a live attack**, after a data breach. Contained the incident, traced the entry route and closed the whole class of weakness. The site stayed up through a Layer-7 DDoS peaking at 700M requests, **∼600M of them blocked**.
- Rebuilt the AWS architecture around one rule: traffic reaches the application only through CloudFront and the WAF. Locked the origin, moved workloads to ECS Fargate behind a private load balancer, and encrypted passport scans with AES-256-GCM before they are written to S3. Moved the platform off end-of-life Node 20; the image build now refuses an unpatched runtime.
- **Led the three-engineer team** after the recovery: built the admin suite staff use to process applications (call queues, auto-assignment, productivity reports, GDPR erasure and retention), integrated Adyen, MamoPay and PayPal, and built staff sign-in with authenticator-app two-factor and passkeys (WebAuthn). **Revenue passed $1M.**
- Launched a second brand, Crown Visa, on the same platform: each brand ships as its own build, which fails if the other brand's text leaks in, and payments, emails and refunds follow the brand and domain a customer used.
- **Co-led WorldVisaHub**, a multi-country visa platform where supervisors build each country's application form themselves. Made published form versions immutable and pinned to in-flight orders, so an edit never changes an application in progress.
- Designed its document pipeline with **no long-lived cloud keys**: uploads go straight to S3 on short-lived STS credentials, files are checked by content rather than extension, encrypted with KMS, and personal data is scrubbed from logs. Provisioned staging in Terraform, with keyless OIDC deploys gated on Brakeman and bundler-audit.
- Building the NestJS and OPC UA backend of a live industrial grain-aeration SCADA platform, in which one service holds every site's PLC session: connections open in stages with jittered retries, every node is re-read after an outage, and each organisation's admin manages only its own PLCs. Deployed on AWS ECS through Terraform and CI.

Associate Software Engineer Jun 2025 – Dec 2025
OptimaGeeks *Lahore, Pakistan*

- **Led Userbird (now LinklyHQ)**, an analytics and session-replay platform: an embeddable JavaScript SDK streaming pageviews, events and recordings into BigQuery. Added caching and lazy loading that made the dashboard **40% faster**.
- Shipped production apps including a LinkedIn Sales Navigator to CRM extension, OpenAI-powered CRM tooling, WordPress publishing automation, a bot on Cloudflare Workers and a NestJS backend.

Associate Software Developer Mar 2023 – Feb 2025
Webmended (UAE) *Remote*

- **Led mDrive Asia** end to end: storefront, checkout, admin and fulfilment, including Apple GSX and custom payment integrations (Next.js, Django, PostgreSQL). Also built a Django quality-management system and PHP client projects.

LAMP Stack Developer Feb 2022 – Mar 2023
easybew (Canada) *Remote*

Projects

ukvisaportal.com	UK visa portal and admin suite	*Node.js, TypeScript, PostgreSQL, Prisma, Redis, AWS, Cloudflare*	2026
worldvisahub.com	form builder with 24 field types, camera capture	*Rails 8, React 19, PostgreSQL, Sidekiq, AWS*	2026
app.aerationcontrol.com	grain-aeration SCADA	*NestJS, OPC UA, Redis, TimescaleDB, React 19, SSE, AWS*	2026
linklyhq.com	link shortening, analytics and session replay	*React, Vite, BigQuery, Node.js, TypeScript, Cloudflare*	2025
Mirai	ClickUp-style work management	*React, TypeScript, Express, Prisma, PostgreSQL, Redis, Socket.io*	2025

Publications

First author. "AI-Assisted Automated Software Testing and Vulnerability Detection Using Large Language Models." *Spectrum of Engineering Sciences* 4(3), 2026.

Co-author. "Transfer Learning Optimization for Low-Resource Natural Language Processing Using Transformer Architectures." *Research Consortium Archive* 4(2), 2026.

Skills

Build: TypeScript, Node.js (NestJS, Express), React, Next.js, Python (Django), Ruby on Rails, OpenAI API
Data: PostgreSQL, Redis, MongoDB, BigQuery, SQL
Run: AWS (ECS Fargate, RDS, S3, CloudFront, WAF, KMS, IAM/STS), Terraform, Docker, GitHub Actions, Cloudflare
Secure: OAuth 2.0, Microsoft SSO, RBAC, TOTP and WebAuthn passkeys, encryption at rest, WAF and DDoS response, GDPR

Education

Government College University (GCU) Lahore Nov 2021 – May 2025
Bachelor of Science in Computer Science (BSCS), CGPA 3.09 *Lahore, Pakistan*